

Unique continuation properties of the nonlinear Schrödinger equation

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Consider the unique continuation problem for the nonlinear Schrödinger (NLS) equation

$$iu_t + u_{xx} + \alpha|u|^2u = 0, \quad -\infty < x, t < +\infty.$$

By using the inverse scattering transform and some results from the Hardy function theory, we prove that if $u \in C(R; H^1(R))$ is a solution of the NLS equation, then it cannot have compact support at two different moments unless it vanishes identically. In addition, it is shown under certain conditions that if u is a solution of the NLS equation, then u vanishes identically if it vanishes on two horizontal half lines in the x - t space. This implies that the solution u must vanish everywhere if it vanishes in an open subset in the x - t space.

1. Introduction

In this paper, we consider unique continuation problems for the nonlinear Schrödinger equation

$$iu_t + u_{xx} + \alpha|u|^2u = 0, \quad -\infty < x, t < +\infty,$$

where α is a real constant and the subscripts t and x denote partial differentiation.

We show that if $u(x, t) \in C(R; H^1(R))$ is a solution of the NLS equation, then it cannot have compact support at two different moments unless it vanishes identically. In addition, we prove under certain conditions that if $u(x, t)$ is a solution of the NLS equation, then $u(x, t)$ must vanish identically if it vanishes on two horizontal half lines in the x - t space. As a consequence, if $u(x, t)$ is a solution of the NLS equation and vanishes on an open subset in the x - t space, then it vanishes everywhere. These results are similar to those for the Korteweg–de Vries (KdV) equation obtained in our previous paper [19].

Usually, an evolution equation is said to have finite propagation speed if its solution has compact support for all time provided the initial data have compact support. Otherwise, the equation is said to have infinite propagation speed. It is well known that some nonlinear evolution equations and their linear counterparts may possess different propagation speed. For instance, the heat equation

$$u_t(x, t) = u_{xx}(x, t), \quad x \in R,$$

has infinite propagation speed. But the nonlinear equation

$$u_t = u_{xx} + cu^p, \quad x \in R,$$

with $c < 0$ and $0 < p < 1$, has finite propagation speed (see [12]). Evans and Knerr even showed that the support of some solutions to certain parabolic equations and variational inequalities will shrink instantaneously to be compact, although their initial data do not have compact support (see [5]).

The unique continuation results established in this paper clearly show that the NLS equation possesses infinite propagation speed (so does its linear counterpart). Moreover, if the initial data of a nontrivial solution to the NLS equation have compact support, then the support will spread out immediately and never shrink back to be compact.

The study of unique continuation properties for partial differential operators began with Carleman's work [2], in which he proved that the operator $L = \Delta + v(x)$ in R^2 , with $v \in L_{loc}^\infty(R^2)$, has the strong unique continuation property, i.e. $u \equiv 0$ is the only solution to $Lu = 0$ which has infinite zero at a point in R^2 . Since then, many unique continuation results have been obtained for elliptic operators (cf. [7, 9, 15] as well as many references contained in these papers). As regards evolution equations, Mizohata [13], and Saut and Scheurer [16] showed that a class of second-order parabolic operators L with variable coefficients, defined on a connected open set $\Omega \subset R_t \times R_x^n$, has the unique continuation property that every solution u of $Lu = 0$ which vanishes on an open subset Q of Ω vanishes in the horizontal component of Q in Ω , i.e. in $\{(x, t) \in \Omega; \text{there exists } x_1, (x_1, t) \in Q\}$. Some higher-order parabolic operators and some dispersive operators in one space dimension are also considered by Saut and Scheurer in [16]. Kenig and Sogge [11] considered the Schrödinger operator. They proved that if $u(x, t) \in H^{2(n+1)/(n+4)}(R^{n+1})$ satisfies the differential inequality

$$\left| \left(i \frac{\partial}{\partial t} + \Delta \right) u(x, t) \right| \leq |v(x, t)u(x, t)|$$

for some $v(x, t) \in L^{(n+2)/2}(R^{n+1})$, then $u(x, t)$ must vanish identically if it vanishes in a half space in R^{n+1} . Similar results are obtained for the hyperbolic operator $\partial_{tt} - \Delta$ by Kenig, Ruize and Sogge [10].

The major method for studying unique continuation properties of partial differential operators is to establish the corresponding Carleman-type inequality, which originates from Carleman's pioneering work [2]. However, our approach is quite different and is mainly based on some basic facts of Hardy functions. The key ingredient of our approach is an estimate of Hardy functions in the class of H_+^2 . More precisely, let $z(t)$ be a continuous curve in the upper half complex plane with $z(t) \rightarrow \infty$ as $t \rightarrow \infty$. Then, for any $0 \neq h(z) \in H_+^2$, there exist a $\delta > 0$ and a sequence of $\{t_n\}$ with $t_n \rightarrow +\infty$ such that

$$|h(z(t_n))| \geq \exp \left\{ -\delta \frac{|z(t_n)|^2}{\text{Im } z(t_n)} \right\}$$

for $n = 1, 2, \dots$. This estimate has been proved in our earlier papers [19, 20]. Based on this estimate, it is quite elementary to obtain the unique continuation properties described above for the corresponding linear Schrödinger equation. Then the unique continuation properties for the nonlinear Schrödinger equation are established by using an inverse scattering transformation.

This paper is organised as follows. In Section 2, we recall some basic definitions and facts of Hardy functions. An estimate of the decaying rate of Hardy functions is also provided. Then, by virtue of the Fourier transform, we show some unique continuation properties for the linear Schrödinger equation. In Section 3, we cite some results concerning the direct and inverse scattering for the generalised Zakharov–Schabat eigenvalue problem from [1, 18], and provide a sketch of the inverse scattering transform to the NLS equation. This will play an important role in the proof of our main results. In Section 4, we prove our main continuation results for the NLS equation. We see that the inverse scattering transformation in the nonlinear problem plays the same role as that of the Fourier transform in the linear problem.

2. Linear Schrödinger equation

We use the following notation to denote the Fourier transform and the inverse Fourier transform:

$$\hat{f}(y) = \frac{1}{\pi} \int_{-\infty}^{+\infty} e^{2ixy} f(x) dx$$

and

$$\check{f}(x) = \frac{1}{\pi} \int_{-\infty}^{+\infty} e^{-2ixy} f(y) dy.$$

Let H_+^2 be the Hardy space of functions $h(k)$ analytic in $\text{Im } k > 0$ with

$$\sup_{b>0} \int_{-\infty}^{+\infty} |h(a+ib)|^2 da < +\infty. \quad (2.1)$$

Such a function $h(k)$ assumes boundary values

$$\lim_{\varepsilon \rightarrow 0^+} h(a + i\varepsilon)$$

in $L^2(\mathbb{R})$. It is well known [3] that

$$H_+^2 = \{h(k) \in L^2(\mathbb{R}); \text{supp } \hat{h} \subset (-\infty, 0)\}. \quad (2.2)$$

Similarly, H_-^2 denotes the Hardy space of functions analytic in $\text{Im } k < 0$ with

$$\sup_{b<0} \int_{-\infty}^{+\infty} |h(a+ib)|^2 da < \infty$$

and

$$H_-^2 = \{h \in L^2(\mathbb{R}); \text{supp } \hat{h} \subset (0, \infty)\}.$$

From (2.1) and (2.2), we have the following lemma:

LEMMA 2.1. *If $h(k) \in L^2(\mathbb{R})$ and the support of $\check{h} \subset (0, \infty)$, then $h(k)$ can be extended to the upper half complex plane as an analytic function and*

$$\sup_{b>0} \int_{-\infty}^{\infty} |h(a+ib)|^2 da < +\infty.$$

The following lemma, which gives a lower bound estimate for any $h(k) \in H_+^2$, will be a key ingredient in the proof of our main results.

LEMMA 2.2. *If $0 \neq h \in H_+^2$, then for any continuous curve*

$$s(\tau) = u(\tau) + iv(\tau), \quad 0 \leq \tau < +\infty,$$

in the upper half complex plane with

$$\lim_{\tau \rightarrow +\infty} s(\tau) = \infty,$$

there exists a $\delta > 0$ and a sequence of $\{\tau_n\}$ with

$$\lim_{n \rightarrow +\infty} \tau_n = +\infty$$

such that

$$|h(s(\tau_n))| \geq \exp \left\{ -\delta \frac{u^2(\tau_n) + v^2(\tau_n)}{v(\tau_n)} \right\}$$

for $n = 1, 2, 3, \dots$

The complete proof of this lemma can be found in [20].

Now we consider the one-dimensional linear Schrödinger equation

$$u_t(x, t) - iu_{xx}(x, t) + \alpha u(x, t) = 0, \quad -\infty < x, t < +\infty, \quad (2.3)$$

where α is a complex constant. Formally, by using the Fourier transform, we have

$$u(x, t) = \frac{1}{\pi} \int_{-\infty}^{+\infty} e^{2ikx} e^{(-ik^2 - \alpha)t} \check{q}(k) dk, \quad (2.4)$$

where $q(x) = u(x, 0)$. If $q \in L^2(\mathbb{R})$, the $u(x, t)$ defined in (2.4) belonging to $L^2(\mathbb{R})$ for any t is called a mild solution of (2.3). If q is smooth enough, then the mild solution $u(x, t)$ is a classical solution of (2.3).

THEOREM 2.3. *Let $u(x, t)$ be a mild solution of (2.3). If there exists $t_1 < t_2$ such that*

$$\text{supp } u(\cdot, t_j) \subset (-\infty, \beta), \quad j = 1, 2, \quad (2.5)$$

or

$$\text{supp } u(\cdot, t_j) \subset (\beta, \infty), \quad j = 1, 2, \quad (2.6)$$

for some $\beta \in \mathbb{R}$, then

$$u(x, t) \equiv 0, \quad -\infty < x, t < +\infty.$$

Proof. Without loss of generality, we assume that $t_1 = 0$, $\alpha = 0$ and $u(x, 0) = q(x)$. It suffices to prove the case that u satisfies assumption (2.5).

As a consequence of (2.4) and (2.5),

$$\int_{-\infty}^{+\infty} e^{2ikx} e^{ik^2 t_2} \check{q}(k) dk = 0, \quad \text{for } x \in (0, \infty).$$

In addition, since $e^{ik^2 t_2} \check{q}(k) \in L^2(\mathbb{R})$, we have $e^{ik^2 t_2} \check{q}(k) \in H_+^2$ and

$$\sup_{b > 0} \int_{-\infty}^{+\infty} |e^{i(a+ib)^2 t_2} \check{q}(a+ib)|^2 da < \infty. \quad (2.7)$$

On the other hand,

$$\begin{aligned}
 I(b) &\equiv \int_{-\infty}^{+\infty} |e^{i(a+ib)^2 t_2} \check{q}(a+ib)|^2 da \\
 &= \int_{-\infty}^{\infty} e^{-4abt_2} |\check{q}(a+ib)|^2 da \\
 &\geq \int_{-(b+1)}^{-b} e^{-4abt_2} |\check{q}(a+ib)|^2 da \\
 &\geq e^{2b^2 t_2} \int_{-(b+1)}^{-b} |\check{q}(a+ib)|^2 da \\
 &\geq e^{4b^2 t_2} \left| \int_{-(b+1)}^{-b} \int_{-\infty}^0 q(x) e^{-2i(a+ib)x} dx da \right|^2 \\
 &= e^{4b^2 t_2} \left| \int_{-\infty}^0 q(x) \left(\int_{-(b+1)}^{-b} e^{-2i(a+ib)x} da \right) dx \right|^2 \\
 &= e^{4b^2 t_2} \left| \int_{-\infty}^0 e^{2bx(1+i)} p(x) dx \right|^2,
 \end{aligned}$$

where

$$p(x) = \frac{e^{2ix} - 1}{2ix} q(x).$$

From assumption (2.7), we have $p \in L^2(R)$ and $\text{supp } p \subset (-\infty, 0)$. Thus,

$$h(z) = \int_{-\infty}^{+\infty} e^{-2izx} p(x) dx \in H_+^2.$$

Note that

$$h\left(\frac{b}{2} e^{\frac{\pi}{2} i}\right) = \int_{-\infty}^0 e^{2b(1+i)x} p(x) dx.$$

According to Lemma 2.1, if $p \neq 0$, there exists a sequence of $\{b_n\}$ with $b_n \rightarrow \infty$ such that

$$\left| \int_{-\infty}^0 e^{2b_n x(1+i)} p(x) dx \right|^2 \geq e^{-\delta b_n}$$

for some $\delta > 0$ and $n = 1, 2, \dots$. Thus,

$$I(b_n) \geq e^{4b_n^2 t_2} e^{-\delta b_n}$$

for $n = 1, 2, \dots$. Letting $n \rightarrow +\infty$, we obtain

$$\lim_{n \rightarrow \infty} I(b_n) = +\infty,$$

which contradicts (2.7). So we must have $p(x) = 0$. As a consequence of uniqueness,

$$u(x, t) = 0, \quad \text{for any } x \in R \text{ and } t \in R.$$

The proof is complete. $\square$

COROLLARY 2.4. *Let $u(x, t)$ be a mild solution of (2.3). Then u vanishes identically if it vanishes on an open subset in x - t space.*

Proof. Without loss of generality, we assume that

$$u(x, t) = 0, \quad x \in (a, b), \quad \text{and} \quad t \in (t_1, t_2)$$

for some $a, b \in R$ and $t_1, t_2 \in R$. Define

$$\tilde{u}(x, t) = \begin{cases} u(x, t), & \text{if } x \geq c, t \in (t_1, t_2), \\ 0, & \text{if } x < c, t \in (t_1, t_2), \end{cases}$$

where $c = (a + b)/2$. Then $\tilde{u}(x, t)$ is a mild solution of (2.3) for $t_1 < t < t_2$. By Theorem 2.3, $\tilde{u}(x, t)$ vanishes for $t_1 < t < t_2$. Hence $u(x, t)$ vanishes on the strip $(-\infty, c) \otimes (t_1, t_2)$ and therefore vanishes everywhere by Theorem 2.3. The proof is complete. $\square$

3. Direct and inverse scattering problem

Consider the generalised Zakharov-Schabat eigenvalue problem

$$\begin{cases} v_{1x} + ikv_1 = q(x)v_2, \\ v_{2x} - ikv_2 = r(x)v_1 \end{cases} \quad (3.1)$$

on the real line R , where the parameter k plays the role of eigenvalue and the potentials $q(x)$ and $r(x)$ are complex valued functions. We assume that $q(x), r(x) \in L^1(R)$ and $r_x(x), q_x(x) \in L^1(R)$ and define the linear independent solutions to (3.1) which have the following asymptotic values:

$$\begin{aligned} \psi(x, k) &\rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{ikx}, & x \rightarrow +\infty, \\ \bar{\psi}(x, k) &\rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-ikx}, & x \rightarrow +\infty, \\ \varphi(x, k) &\rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-ikx}, & x \rightarrow -\infty, \\ \bar{\varphi}(x, k) &\rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{ikx}, & x \rightarrow -\infty. \end{aligned}$$

These solutions $\varphi, \psi, \bar{\varphi}$ and $\bar{\psi}$ are called *Jost solutions* to (3.1). For every $x \in R$, $\varphi(x, k), \psi(x, k)$ are continuous with respect to k on $\text{Im } k \geq 0$ and analytic with respect to k in $\text{Im } k > 0$. The analogous properties hold for $\bar{\psi}(x, k)$ and $\bar{\varphi}(x, k)$ in $\text{Im } k \leq 0$.

The Jost solutions ψ and $\bar{\psi}$ form the basis, so that

$$\begin{aligned} \varphi(x, k) &= a(k)\bar{\psi}(x, k) + b(k)\psi(x, k), \\ p\bar{h}i(x, k) &= \bar{b}(k)\bar{\psi}(x, k) + \bar{a}(k)\psi(x, k). \end{aligned}$$

The coefficients $a(k)$, $b(k)$, $\bar{a}(k)$ and $\bar{b}(k)$ are given by the Wronskians of the Jost solutions φ , $\bar{\varphi}$, ψ , $\bar{\psi}$:

$$a = W(\varphi, \psi), \quad b = W(\bar{\psi}, \varphi),$$

$$\bar{a} = W(\bar{\psi}, \bar{\varphi}), \quad \bar{b} = W(\bar{\varphi}, \psi),$$

where $W(u, v) = v_1 v_2 - v_2 v_1$. The functions $a(k)$, $\bar{a}(k)$, $b(k)$, $\bar{b}(k)$, which are called *scattering functions*, are uniquely defined and continuous in $\text{Im } k \geq 0$, $\text{Im } k \leq 0$, $\text{Im } k = 0$, $\text{Im } k = 0$ respectively. Moreover, $a(k)$, $\bar{a}(k)$ are analytic in $\text{Im } k > 0$, $\text{Im } k < 0$, respectively.

The following propositions are cited from [1] and [18], except Propositions 3.4 and 3.5.

PROPOSITION 3.1. *The Jost functions φ , ψ , $\bar{\varphi}$, and $\bar{\psi}$ have the uniformly valid asymptotic expansions as follows:*

$$\varphi(x, k)e^{ikx} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{2ki} \begin{pmatrix} \int_{-\infty}^x r(y)q(y) dy \\ r(x) \end{pmatrix} + o(|k|^{-1}),$$

$$\psi(x, k)e^{-ikx} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \frac{2}{2ki} \begin{pmatrix} q(x) \\ - \int_x^{+\infty} r(y)q(y) dy \end{pmatrix} + o(|k|^{-1}),$$

$$\bar{\psi}(x, k)e^{ikx} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{2ki} \begin{pmatrix} \int_{+\infty}^x r(y)q(y) dy \\ -r(x) \end{pmatrix} + o(|k|^{-1}),$$

$$\bar{\varphi}(x, k)e^{-ikx} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \frac{1}{2ki} \begin{pmatrix} q(x) \\ \int_x^{+\infty} r(y)q(y) dy \end{pmatrix} + o(|k|^{-1}),$$

as $k \rightarrow \infty$ within their domains of definition.

PROPOSITION 3.2. *The scattering functions $a(k)$, $\bar{a}(k)$, $b(k)$ and $\bar{b}(k)$ may be expressed by Jost solutions as follows:*

$$a(k) = 1 + \int_{-\infty}^{+\infty} q(y)e^{iky}\varphi_2(y, k) dy,$$

$$\bar{a}(k) = 1 + \int_{-\infty}^{+\infty} r(y)e^{-iky}\bar{\varphi}_1(y, k) dy,$$

$$b(k) = \int_{-\infty}^{+\infty} r(y)e^{-iky}\varphi_1(y, k) dy,$$

and

$$\bar{b}(k) = \int_{-\infty}^{+\infty} q(y)e^{iky}\bar{\varphi}_2(y, k) dy.$$

In addition, $k = k_0$ is an eigenvalue or a spectral singularity to eigenvalue problem (3.1) if and only if k_0 is a zero of $a(k)$ or $\bar{a}(k)$.

PROPOSITION 3.3. *The scattering functions have the following asymptotic behaviour:*

$$a(k) = 1 - \frac{1}{2ki} \int_{-\infty}^{+\infty} r(y)q(y) dy + o(|k|^{-1}),$$

$$\bar{a}(k) = 1 - \frac{1}{2ki} \int_{-\infty}^{+\infty} r(y)q(y) dy + o(|k|^{-1}),$$

and

$$b(k) = o(|k|^{-1}), \quad \bar{b}(k) = o(|k|^{-1})$$

as $|k| \rightarrow \infty$ within their domains of definition.

PROPOSITION 3.4 (trace formula). *If $a(k) \neq 0$ for $\text{Im } k \geq 0$ and $\bar{a}(k) \neq 0$ for $\text{Im } k \leq 0$, then*

$$r(x) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{b(k)}{a(k)} \psi_2(x, k) e^{ikx} dk$$

and

$$q(x) = -\frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{\bar{b}(k)}{\bar{a}(k)} \bar{\psi}_1(x, k) e^{-ikx} dk.$$

Proof. Since both $a(k)$ and $\bar{a}(k)$ have no zeros, according to [1], we have

$$\psi(x, k) e^{-ikx} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \frac{1}{2\pi i} \int_{-\infty}^{+\infty} \frac{\bar{b}(\lambda)}{\bar{a}(\lambda)} \bar{\psi}(x, \lambda) e^{-i\lambda x} \frac{1}{\lambda - k} d\lambda$$

and

$$\bar{\psi}(x, k) e^{ikx} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{2\pi i} \int_{-\infty}^{+\infty} \frac{b(\lambda)}{a(\lambda)} \psi(x, \lambda) e^{i\lambda x} \frac{1}{\lambda - k} d\lambda.$$

Thus

$$\psi(x, k) e^{-ikx} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \frac{1}{2k\pi i} \int_{-\infty}^{+\infty} \frac{\bar{b}(\lambda)}{\bar{a}(\lambda)} \bar{\psi}(x, \lambda) e^{-i\lambda x} d\lambda + o\left(\frac{1}{k^2}\right)$$

and

$$\bar{\psi}(x, k) e^{ikx} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{2k\pi i} \int_{-\infty}^{+\infty} \frac{b(\lambda)}{a(\lambda)} \psi(x, \lambda) e^{i\lambda x} d\lambda + o\left(\frac{1}{k^2}\right),$$

which, together with Proposition 3.1, imply that

$$q(x) = -\frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{\bar{b}(\lambda)}{\bar{a}(\lambda)} \bar{\psi}_1(x, \lambda) e^{-i\lambda x} d\lambda$$

and

$$r(x) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{a(\lambda)}{b(\lambda)} \psi_2(x, \lambda) e^{i\lambda x} d\lambda.$$

The proof is complete. $\square$

PROPOSITION 3.5. *Under the hypotheses of Proposition 3.4, if $\text{supp } r(x) \subset (-\infty, 0)$, then*

$$\int_{-\infty}^{+\infty} \frac{b(\lambda)}{a(\lambda)} e^{2i\lambda x} d\lambda = 0, \quad \text{for } x \geq 0. \quad (3.2)$$

Similarly, if $\text{supp } q(x) \subset (-\infty, 0)$, then

$$\int_{-\infty}^{+\infty} \frac{\bar{b}(\lambda)}{\bar{a}(\lambda)} e^{-2i\lambda x} d\lambda = 0, \quad \text{for } x \geq 0. \quad (3.3)$$

Proof. If $\text{supp } r(x) \subset (-\infty, 0)$, we have

$$\psi_2(x, k) = e^{ikx}, \quad \text{for } x \geq 0,$$

since ψ_2 is a unique solution of

$$\frac{d}{dx} \psi_2(x) - ik\psi_2(x) = 0, \quad x \in (0, +\infty),$$

with

$$\lim_{x \rightarrow +\infty} \psi_2(x) e^{-ikx} = 1.$$

By virtue of the trace formula in Proposition 3.4, we have

$$\int_{-\infty}^{+\infty} \frac{b(\lambda)}{a(\lambda)} e^{2i\lambda x} d\lambda = 0 \quad \text{for } x \geq 0.$$

(3.3) follows from the same argument. The proof is complete. $\square$

PROPOSITION 3.6. *If $r(x) = q^*(x)$ (complex conjugate of $q(x)$), then*

$$\bar{a}(k) = a^*(k^*), \quad \bar{b}(k) = -b^*(k^*)$$

and

$$|a(k)| \geq 1, \quad \text{for } \text{Im } k \geq 0.$$

PROPOSITION 3.7. *If $r(x) = -q^*(x)$, then*

$$\bar{a}(k) = a^*(k^*), \quad \bar{b}(k) = b^*(k^*)$$

and

$$|a(k)|^2 + |b(k)|^2 = 1, \quad \text{for real } k.$$

In addition, if

$$\int_{-\infty}^{+\infty} |r(x)| dx < \ln(2 + \sqrt{3}),$$

then

$$|a(k)| > 0 \quad \text{for } \operatorname{Im} k \geq 0.$$

Finally, we consider the initial value problem for the NLS equation:

$$iy_t(x, t) + y_{xx}(x, t) \pm 2|y(x, t)|^2 y(x, t) = 0, \quad y(x, 0) = y_0(x), \quad (3.4)$$

for $-\infty < x, t < +\infty$.

PROPOSITION 3.8. *If the scattering functions corresponding to the potential $q(x) = y_0(x)$ and $r(x) = \mp y_0^*(x)$ for eigenvalue problem (3.1) are $a(k)$ and $b(k)$, then the scattering functions corresponding to the potential $q(x) = y(x, t)$ and $r(x) = \mp^* y(x, t)$ (considering t as a parameter) are given by*

$$a(k, t) = a(k), \quad b(k, t) = b(k)e^{4ik^2t} \quad \text{for any } t,$$

where $y(x, t)$ is the solution of (3.4) corresponding to the initial data $y_0(x)$.

4. Nonlinear Schrödinger equation

Consider the initial value problem for the NLS equation:

$$iu_t + u_{xx} + \alpha|u|^2u = 0, \quad u(x, 0) = q(x), \quad (4.1)$$

where $x \in \mathbb{R}$, $t \in \mathbb{R}$ and α is a real constant. It is known [17] that if $q \in H^1(\mathbb{R})$, then (4.1) admits a unique solution $u \in C(\mathbb{R}; H^1(\mathbb{R}))$.

Considering the unique continuation problem of the equation (4.1), we first assume that $\alpha < 0$ in (4.1). In this case, the solutions of equation (4.1) possess the following unique continuation properties.

THEOREM 4.1. *Let $u \in C(\mathbb{R}; H^1(\mathbb{R}))$ be a solution of the NLS equation (4.1). If there exist $t_1 < t_2$ such that $u(\cdot, t_1) \in L^1(\mathbb{R})$, $u_x(\cdot, t_1) \in L^1(\mathbb{R})$ and*

$$\operatorname{supp} u(\cdot, t_j) \subset (-\infty, \beta), \quad j = 1, 2, \quad (4.2)$$

or

$$\operatorname{supp} u(\cdot, t_j) \subset (\beta, +\infty), \quad j = 1, 2, \quad (4.3)$$

for some $\beta \in \mathbb{R}$, then

$$u(x, t) \equiv 0, \quad -\infty < x, t < +\infty.$$

Proof. Without loss of generality, we assume $\alpha = -2$, $t_1 = 0$ and $\beta = 0$. It suffices to consider only the case wherein u satisfies assumption (4.2).

Suppose $a(k)$ and $b(k)$ are scattering functions of the generalised Zakharov–Schabat eigenvalue problem (3.1) corresponding to $q(x)$ and $q^*(x)$. Then by Proposition 3.8 the scattering functions corresponding to the solutions $u(x, t)$ and $u^*(x, t)$ are

$$a(k, t) = a(k), \quad b(k, t) = b(k)e^{4ik^2t},$$

for any $t \in \mathbb{R}$. Thus we have, by virtue of the trace formula in Proposition 3.4,

$$\begin{aligned} u^*(x, t) &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{b(\lambda, t)}{a(\lambda, t)} \psi_2(x, \lambda, t) e^{i\lambda x} d\lambda \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{b(\lambda)}{a(\lambda)} e^{4i\lambda^2 t} \psi_2(x, \lambda, t) e^{i\lambda x} d\lambda, \end{aligned}$$

where $\psi_2(x, \lambda, t)$ is the Jost solution to (3.1) corresponding to the potential $u(x, t)$ and $u^*(x, t)$. It follows from assumption (4.2) and Proposition 3.5 that

$$\int_{-\infty}^{+\infty} \frac{b(\lambda)}{a(\lambda)} e^{4i\lambda^2 t_2} e^{2i\lambda x} d\lambda = 0, \quad \text{for } x \geq 0. \quad (4.4)$$

According to Proposition 3.3,

$$|a(\lambda)| \rightarrow 1, \quad |b(\lambda)| = o\left(\frac{1}{|\lambda|}\right) \quad \text{as } |\lambda| \rightarrow \infty.$$

Hence

$$\frac{b(\lambda)}{a(\lambda)} e^{4i\lambda^2 t_2} \in H_+^2. \quad (4.5)$$

By definition of the Hardy function,

$$\sup_{d>0} \int_{-\infty}^{+\infty} \left| \frac{b(c+id)}{a(c+id)} \right|^2 |e^{4i(c+id)^2 t_2}|^2 dc < \infty.$$

On the other hand, letting

$$I(d) = \int_{-\infty}^{+\infty} \left| \frac{b(c+id)}{a(c+id)} \right|^2 |e^{2i(c+id)^2 t_2}|^2 dc,$$

we have

$$\begin{aligned} I(d) &= \int_{-\infty}^{+\infty} e^{-16cdt_2} \left| \frac{b(c+id)}{a(c+id)} \right|^2 dc \\ &\geq \int_{-d-1}^{-d} e^{-16cdt_2} \left| \frac{b(c+id)}{a(c+id)} \right|^2 dc \\ &\geq e^{16d^2 t_2} \int_{-d-1}^{-d} \left| \frac{b(c+id)}{a(c+id)} \right|^2 dc. \end{aligned}$$

From Propositions 3.1 and 3.2, we have

$$\varphi_1(y, k) = e^{-iky} + o\left(\frac{1}{|k|}\right), \quad \text{for } \operatorname{Im} k \geq 0,$$

uniformly about $y \in \mathbb{R}$, and

$$a(k) = 1 + o\left(\frac{1}{|k|}\right), \quad \text{for } \operatorname{Im} k \geq 0.$$

In addition, by Proposition 3.2,

$$\begin{aligned} b(k) &= \int_{-\infty}^{+\infty} r(y) e^{-iky} \varphi_1(y) dy \\ &= \int_{-\infty}^{+\infty} r(y) e^{-2iky} dy + o\left(\frac{1}{|k|}\right). \end{aligned}$$

Hence there exists a constant C such that

$$\begin{aligned} I(d) &\geq C e^{16d^2 t_2} \int_{-d-1}^{-d} \left| \int_{-\infty}^{+\infty} r(y) e^{-2i(c+id)y} dy \right|^2 dc \\ &\geq C e^{16d^2 t_2} \left| \int_{-\infty}^{+\infty} r(y) e^{2dy} \left(\int_{-d-1}^{-d} e^{-2icy} dc \right) dy \right|^2 \\ &= C e^{16d^2 t_2} \left| \int_{-\infty}^{+\infty} h(y) e^{2dy(1+i)} dy \right|^2, \end{aligned}$$

where

$$h(y) = \frac{e^{2iy} - 1}{2iy} r(y).$$

Note that

$$\begin{aligned} \int_{-\infty}^{+\infty} h(y) e^{2dy(1+i)} dy &= \int_{-\infty}^{+\infty} e^{-2idy(-1+i)} h(y) dy \\ &= \hat{h}((-1+i)d). \end{aligned}$$

We have $\hat{h} \in H_+^2$ since $\text{supp } h(y) \subset (-\infty, 0)$ by our assumption. If $h \neq 0$, according to Lemma 2.1, there exists a sequence of $\{d_n\}$ with $d_n \rightarrow \infty$ such that

$$|\hat{h}((-1+i)d_n)| \geq e^{-\delta d_n}$$

for $n = 1, 2, \dots$. As a result,

$$I(d_n) \geq C e^{16d_n^2 t_2} e^{-2\delta d_n}.$$

Hence

$$\lim_{n \rightarrow \infty} I(d_n) = +\infty,$$

which contradicts (4.5). Thus we must have $h = 0$ as a consequence of uniqueness,

$$u(x, t) \equiv 0, \quad -\infty < x, t < +\infty.$$

The proof is complete. $\square$

From Theorem 4.1, we immediately have the next theorem:

THEOREM 4.2. *Let $u \in C(R, H^1(R))$ be a solution to (4.1). Then the solution $u(x, t)$ cannot have compact support at two different times unless it vanishes identically.*

REMARK 4.3. Theorem 4.2 implies that any information carried by the initial state will be transmitted to everywhere immediately. In other words, the dispersive waves described by the NLS equation have infinite speed of propagation.

Let $\sigma(x) = (1 + |x|^2)^{\frac{1}{2}}$ and define a Hilbert space, for any $s \in \mathbb{R}$ and positive integer n ,

$$H_s^n(R) = \left\{ f(x) \in H^n(R); \left(\sum_{j=0}^n \|\sigma^s f^{(j)}\|_{L^2(R)}^2 \right)^{\frac{1}{2}} < \infty \right\}$$

endowed with the norm

$$\|u\|_{H_s^n} = \left(\sum_{j=0}^n \|\sigma^s f^{(j)}\|_{L^2(R)}^2 \right)^{\frac{1}{2}}.$$

Let

$$S_s^n(R) = H_0^n(R) \cap H_s^n(R).$$

Then

$$S_s^n(R) \subset [H_s^0(R), H_0^n(R)]_\theta = H_{(\Gamma - \theta)s}^\theta(R)$$

for any $0 < \theta < 1$ (cf. [17]). It has been proved by Tsutsumi [17] that if $q \in S_s^n(R)$, there exists a unique solution $u(x, t)$ to (4.1) in the class

$$u \in C(R; S_s^n(R)).$$

Clearly, if $q \in S_2^2$, then $q \in L^1(R)$ and hence from Theorem 4.1 we have the following corollaries:

COROLLARY 4.4. Assume $q \in S_2^2(R)$ and $u(x, t)$ be a solution to (4.1). Then u vanishes identically if it vanishes on two different horizontal half lines in the x - t space.

COROLLARY 4.5. Suppose $q \in S_2^2(R)$. If $u(x, t)$ is a solution to (4.1) and vanishes in an open subset in the x - t space, u vanishes everywhere.

COROLLARY 4.6. Assume that $q \in S_2^2(R)$ and that $u(x, t)$ is a solution to (4.1). If there is a $\beta \in \mathbb{R}$ and $t_1 < t_2$ such that

$$u(\beta, t) = 0, \quad u_x(\beta, t) = 0$$

for $t_1 \leq t \leq t_2$, then the solution v vanishes identically.

Corollary 4.4 follows directly from Theorem 4.1. The proofs of Corollaries 4.5 and 4.6 are similar to the proof of Corollary 2.4.

Now we turn to consider the case $\alpha > 0$ in (4.1). A similar proof gives us the following theorem:

THEOREM 4.7. If the initial value $q(x)$ is assumed to satisfy

$$\int_{-\infty}^{+\infty} |q(x)| dx < \sqrt{\frac{2}{\alpha}} \ln(2 + \sqrt{3}), \quad (4.6)$$

then Theorems 4.1–4.2 and Corollaries 4.4–4.6 also hold for the solution of equation (4.1) in the case $\alpha > 0$.

In the above theorem, assumption (4.6) is used to ensure that the scattering function $a(k)$ corresponding to the potential $q(x)$ and $-q^*(x)$ has no zeros. Otherwise, if $a(k)$ has zeros, that is, the generalised Zakharov–Schabat eigenvalue problem (3.1)

has eigenvalues, our agreement in the proof of Theorem 4.1 does not follow since we can no longer use the trace formula (cf. Proposition 3.4). In our previous paper [19] for the KdV equation, we used the Deift–Trubowitz–Crum [3] method to eliminate the eigenvalues. However, the generalised Zakharov–Schabat eigenvalue problem is more complicated than the Schrödinger eigenvalue problem related to the KdV equation. In the case of the KdV equation, there is only a finite number of eigenvalues and each of them is simple. But for the nonlinear Schrödinger equation (4.1), the corresponding generalised Zakharov–Schabat eigenvalue problem may have infinitely many eigenvalues with multiple multiplicity. There is no method similar to the Deift–Trubowitz–Crum method available for the generalised Zakharov–Schabat eigenvalue problem. As a result, we had to put assumption (4.6) in Theorem 4.3. As we have mentioned in the Introduction, the main approach for establishing unique continuation properties is to obtain the corresponding Carleman estimate. However, the unique continuation properties described in Theorems 4.1 and 4.2 are different from those unique continuation properties studied in [2, 7, 10, 13, 16]. It seems to the author that it would be very difficult to establish the unique continuation properties described in Theorems 4.1 and 4.2 by using the Carleman estimate approach, if not impossible. Nevertheless, we believe that Theorem 4.7 should be true without assumption (4.6). How to prove it is an interesting open question, since it demands a new approach to unique continuation problems of partial differential equations.

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